Cross-sectional Prediction of Diabetic Foot Ulcer in Diabetic Patients using Machine Learning with EHR from Canadian Primary Care Clinics

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Introduction

Diabetic foot ulcers (DFU) are a burdensome complication of diabetes with a global prevalence of 6.4% in type 2 diabetes patients[1]. More than half of diabetic ulcers become infected, and approximately 20% of moderate or severe diabetic foot infections lead to amputation.[2] Early identification of high-risk patients is critical for preventing diabetic foot ulcers. Poor glycemic control, indicated by elevated hemoglobin A1C levels (HbA1C) and fasting blood sugar (FBS), is a well-established independent predictor of diabetic foot ulcer[3]. A patient's diabetic medication history serves as a crucial proxy for diabetes severity. Specifically, a prolonged duration of antidiabetic therapy and the progression to a second-line agent reflect a more advanced stage of diabetes and a higher risk of complications[4]. Long-term use of corticosteroids in patients with diabetes hampers wound healing and exacerbates existing hyperglycemia[5,6]. Previous studies have developed diabetic foot ulcer prediction models using logistic regression and machine learning algorithms such as random forests and decision trees[7,8]. Stegge et al. used age, history of neuropathy, and lesion as predictors in a regression model and found a mean AUC of 0.69 [8]. In terms of feature selection, prior risk models mainly used baseline demographic data, foot-related characteristics, and glycemic measurements as predictors[9,10]. However, a significant gap remains in the use of antidiabetic medication history and corticosteroid use in the diabetic foot ulcers prediction model.

To address this gap, this study proposes two main research questions: (1) Can a cross-sectional combination of routine glycemic indicators (HbA1c, FBS) and cumulative medication histories (duration of corticosteroids and antidiabetic agents) predict the occurrence of diabetic foot ulcers? (2) What is the difference in model performance of logistic regression, Gaussian Naïve Bayes (GNB), and XGBoost models in predicting diabetic foot ulcers? The prediction models empower primary care providers to flag high-risk diabetic foot ulcer patients for early prevention and avoid amputation. Furthermore, comparing different models provides guidance on selecting a robust algorithm and on addressing the class imbalance inherent in diabetic foot ulcer data in the real-world diabetes population.

Methods

The current analysis uses a subset of data from the Canadian Primary Care Sentinel Surveillance Network (CPCSSN)[11], a chronic disease surveillance system based on de-identified electronic health records (EHR) from primary care clinics across the country. This sub-dataset includes EHR data entries dating from 1990 to 2015, with 808,921 rows and 171,619 unique patient IDs. Data analysis was completed in Python (version 4.13). We limited our data to patient entries that indicated a diagnosis of diabetes and dropped missing entries of DFU, resulting in 31,297 observations. The following features were selected for DFU prediction model building: corticosteroid and antidiabetic (first- and second-line) medication use duration in days, HbA1C, and FBS.

Data Exploration During data exploration, we inspected our variables of interest for missing values, implausible entries, or outliers based on clinical implausibility. Box plots were used to identify outliers and assess patterns of missing data in the features. To preprocess the data, missing data values for duration of medication variables were replaced with the value 0 to indicate no medication was taken. Multicollinearity between selected variables was assessed using a Seaborn heatmap correlation matrix, where variables with correlation coefficients exceeding 0.70 are suspected to be highly correlated, and one variable would need to be dropped from the model.

To prevent within-patient data leakage between training and test sets, we first grouped observations by patient ID and stratified the data split by DFU status at the patient level into 80% training and 20% test sets. This was to ensure that there were positive instances of the outcome in both splits of data. We conducted hyperparameter tuning using stratified k-fold cross-validation (k = 5) and grid search for all three models. For logistic regression, we tested on C values 0.01, 0.1, 1, 10, and 100; class weight and with or without "balanced" class weight. We also employed L2 (ridge) regularisation to penalize and prevent coefficients from overfitting to noise in the training data. For GNB, we tuned for multiple smoothing values to account for zero probability if a feature's variance results in zero by adding a small value, over 13 logarithmically spaced values ranging from 10^{-12} to 10^{-6}. Finally, hyperparameter tuning was conducted on XGBoost across 25, 50, and 100 n estimators, max depth of 1, 3, 5, and 7, learning rate of 0.05 and 0.1, and minimum sample leaf of 1, 5, and 10. Recall was used as the primary optimisation metric during cross-validation due to class imbalance.

Multiple imputations by chain equations (MICE) were performed to address missing data (Missing at Random) for HbA1c to produce robust estimations of missing values using information from other variables in the data. Imputations were performed after splitting to prevent data leakage.

The distribution of DFU among those with diabetes was 130 (0.4%), representing a notable class imbalance. To mitigate this, oversampling and undersampling techniques were employed, such that the performance of no resampling, random oversampling of the minority class, undersampling of the majority class, and combined using SMOTE with Edited Nearest Neighbours (SMOTEENN) was evaluated against each other. All resampling procedures were implemented within the training set.

Model Implementation We implemented and evaluated two binary classifier models (logistic regression, GNB) and an XGBoost ensemble model to predict the probability that an individual develops DFU, given the selected features of interest. Each has its own advantages, as logistic regression classifiers are easily interpretable but assume that inputs are linearly related to the log odds of the outcome of interest, while Naïve Bayes models require less training data but assume attributes are independent from each other. We also use GNB as our features are continuous data, but it assumes that each feature follows a normal (Gaussian) distribution. Finally, ensemble methods are non-parametric, require minimal data processing, and are flexible in handling missing values and data types. We used XGBoost because it has built-in regularization, is less susceptible to bias, and offers higher accuracy and precision than individual decision tree models.

Using a confusion matrix and classification report, we assessed model performance using weighted average recall and ROC-AUC for training and testing sets. Given that DFU is inherently unbalanced in the real-world diabetes population, with a global prevalence of 6.4% among type 2 diabetics [1], recall (sensitivity) was the primary performance measure, as it assesses the model's ability to detect positive DFU cases and label them positive. Higher recall indicates that the model will be able to predict the risk of outcome correctly. We emphasized particular attention to recall of positive labelled cases as DFU is a rarer outcome among higher-risk diabetic patients, and the cost of false negatives or missing detection of positive cases is greater than false positives in clinical application.

Natural Language Processing (NLP) Pipeline We used search terms combining "machine learning" or "artificial intelligence" with "health equity," "equality," or "diversity" in PubMed on March 20, 2026. We included all 14 diabetes- or DFU-related papers and selected 8 additional papers from the broader search (not limited to diabetes- or DFU-related), for a total of 22 articles. We extracted titles, abstracts, and discussion sections from the selected 22 papers. The text was first converted to lowercase, and special characters were removed. We

then tokenized the cleaned text and applied a customized stop-word list to filter out general academic noise (e.g., “studies,” “method,” “suggest,” “support,” “review”). Finally, lemmatization was applied to reduce words to their base or root forms before generating a word cloud.

Results

Of 234,464 entries with diabetes, it was observed that 203,167 entries DFU’s outcome were missing, which were then removed from analysis. Among the remaining 31,297 observations, 31,167 (99.6%) had no incidence of DFU outcome and 130 (0.4%) had DFU. Although outliers were identified in glycemic biomarkers (A1c, FBS), and duration of corticosteroid, entries were left in the analysis, as they may reflect realistic clinical extremes. Medication use duration is very right-skewed, as a large proportion of observations have a high percentage of missingness and no medication use (Table 1). Overall, there were no highly correlated variables in our dataset. The highest correlation coefficients were between FBS and A1C (0.67) and duration of first-line antidiabetic and second-line antidiabetic medication in days (0.43).

Based on hyperparameter tuning, logistic regression performed the best with C =0.01, L2 regularization, and no class weights. The best variance smoothing hyperparameter for GNB was 10^{-6}. The best hyperparameters for XGBoost were optimized at learning rate of 0.05, maximum depth of 5, 5 minimum samples per leaf, and 100 estimators. We also found the best strategies for logistic and GNB were undersampling the majority class and combining undersampling and oversampling, respectively. However, XGBoost performed best without applying resampling techniques.

While XGBoost and GNB produced higher accuracy and weighted recall, performance is misled by class imbalance, as all positive predictions for DFU from both models were false, leading to a DFU recall of 0.000, and all correct predictions were for negative DFU, as both have much lower macro average recall than logistic regression. Overall, the logistic regression resulted in the best DFU recall (1.00), macro average recall (0.903) and AUC (0.885), with the tradeoff of weaker weighted recall (0.807).

In general, the logistic regression produced a pseudo R-squared of 0.481, with a log-likelihood ratio p-value of $p<0.001$, with duration of second-line antidiabetic medication use (OR 1.58; $p=0.03$), and duration of corticosteroid use (OR 3.21; $p<0.01$) as the only statistically significant predictors (refer to Supplementary Table 1).

Table 1. Descriptive characteristics of selected attributes for modelling.

Variable	Count	Missing (%)	Mean (SD)
A1c	27,308	3,989 (12.75%)	6.70 (0.95)
FBS	31,297	0 (0.00%)	7.00 (1.83)
Duration of corticosteroid use (days)	4,890	26,407 (84.38%)	878.27 (1070.96)
Duration of 1st line antidiabetic (days)	7,868	23,429 (74.86%)	1541.83 (1169.43)
Duration of 2nd line antidiabetic use (days)	3,177	28,120 (89.85%)	1339.70 (929.95)
Diabetic foot ulcer			
Yes	31167 (99.58)		
No	130 (0.42)		

Table 2. Performance metrics and properties for machine learning models predicting DFU on the test sets.

Model	Logistic Regression	Gaussian Naïve Bayes	XGBoost

Optimized hyperparameters	C = 0.01, class weight = None, Model penalty = L2, Model solver = 'lbfgs'	variance smoothing = 1e-6	Learning rate: 0.05 Maximum depth: 5 Minimum samples leaf: 5 n estimators: 100
Best strategy	Undersample Majority	Combined	No resampling
Test ROC–AUC	0.8850	0.7711	0.8265
Accuracy	0.8074	0.888	0.9944
Recall			
No DFU	0.8066	1.0000	0.9988
DFU	1.0000	0.0000	0.0000
Macro	0.9033	0.5000	0.4994
Weighted	0.8074	0.9956	0.9944

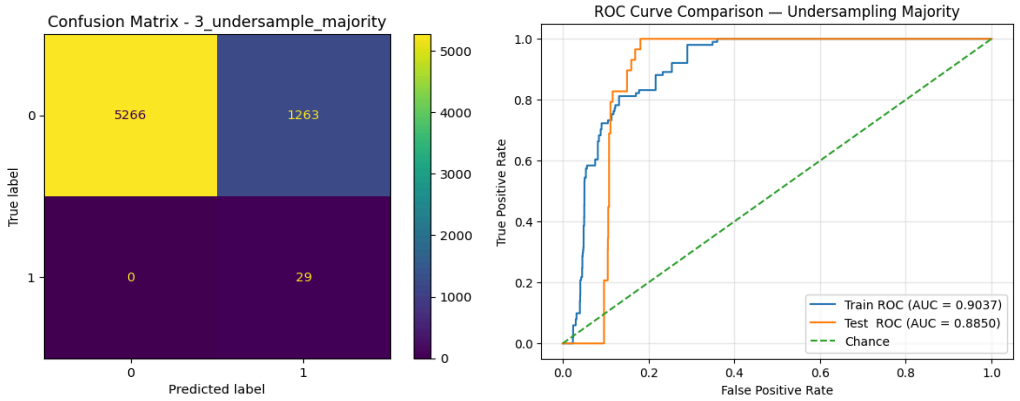

Figure 1. Confusion matrix and ROC-AUC curves for best performing logistic regression model's training and testing sets.

Discussion

In this project, we evaluated whether routine glycemic indicators (HbA1c, FBS) combined with cumulative medication histories can predict diabetic foot ulcer (DFU) occurrence and compared Logistic Regression, Gaussian Naïve Bayes (GNB), and XGBoost models in a highly imbalanced dataset. Our findings suggest that incorporating medication duration variables alongside glycemic biomarkers provides clinically meaningful predictive value, addressing a gap in prior DFU risk models that largely focused on demographic and foot-specific features.

Logistic regression outperformed GNB and XGBoost in DFU recall and macro average recall, indicating the highest relative sensitivity in detecting DFU cases. It is crucial in screening contexts where missed cases carry severe consequences. In clinical practice, logistic regression remains advantageous due to its interpretability and ability to quantify associations between predictors and outcomes.

Limitations However, the extreme class imbalance (0.4% DFU prevalence) and high missingness in medication duration variables present limitations. Although resampling and MICE were applied, further external validation and calibration assessment are needed to confirm generalizability, as high label imbalance may cause high variance and unstable results. In addition, cross-sectional treatment data can risk label leakage between predictors and outcomes, as the temporality of events cannot be ascertained. Although cumulative medication history used in analysis was based on corticosteroids and antidiabetic medications (including first- and second-line antidiabetic medications), and they are not used to treat

DFU[12,13], we found that those who had DFU were 20% and 52% more likely to be prescribed first-line antidiabetic medication and corticosteroids, respectively. In the future, further study could leverage time-to-event machine learning models, sequential models, or temporal framing to address reverse causation by tracking how medication exposure changes over time, such as a sliding window approach [14].

NLP Interpretation The frequent terms in the word cloud (Figure 2), such as "equity," "ethnoracial," "population," "risk," and "fairness," indicate a focus on the risk of ethnoracial bias and the disparity gap. Words like "autonomous," "adherence," and "test" highlight efforts to deploy autonomous AI systems to improve patient adherence and routine testing. Ultimately, these findings suggest that current research emphasizes leveraging AI and equity in workflows to ensure fairness in the ethnoracial diabetes population beyond mere prediction.

Figure 2. Word cloud

Conclusion

The type of the dataset plays a critical role in determining model performance. In this study, Logistic Regression outperformed Gaussian Naive Bayes. This can be attributed to the underlying assumptions of Gaussian Naive Bayes, which include feature independence and normally distributed data. These assumptions were not satisfied in our dataset, thereby limiting the model's performance. Also, class imbalance proved to be a significant challenge that affects model's performance. Despite applying techniques to mitigate this issue, model performance remained affected. In particular, XGBoost exhibited a tendency to misclassify a substantial portion of the minority class (positive DFU cases) as negative. This behavior is largely due to the severe imbalance in class distribution, where positive cases represent only a small fraction of the overall dataset.

Future work should focus on enhancing both the robustness and clinical relevance of the developed models. Incorporating a broader range of clinically significant risk factors including longitudinal patient data and lifestyle indicators which may improve predictive performance and provide a more comprehensive understanding of disease progression. Further validation is essential to ensure generalizability and reliability. Future studies should evaluate model performance using independent cohorts and diverse patient populations to confirm applicability across different clinical settings.

Ultimately, with continued refinement and validation, these models have the potential to be integrated into clinical decision support systems. This would enable early identification of high-risk patients and facilitate timely, targeted preventive care strategies within primary care environments.

Integrating cumulative medication history with routine glycemic indicators enables a proactive, EHR-driven approach to early DFU identification. It empowers primary care providers to identify and intervene with high-risk patients early to prevent severe DFU complications like amputations.

- **Individual Contributions:** Chieh En Chen - introduction/NLP/coding/discussion; Jessie Tian - methods/results/XGBoot/coding; Owusu Boakye Gideon - Logistic regression and GNB/coding/conclusion
- **Code:** https://github.com/GIDEonBoakye/Project-HAD5016

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Supplementary Table 1.

Variable	Coefficient	Odds Ratio	P-value	SD of predictor	% Change Odds per 1 SD	OR per Year (365 days)
A1c	0.246	1.279	0.364	0.950	27.9%	1.297
FBS	0.462	1.587	0.07	1.820	58.7%	1.290
Duration of corticosteroid (days)	1.166	3.210	<0.001	502.46	221.0%	2.333
Duration of 1st-line antidiabetic (days)	-0.341	0.711	0.182	868.24	28.9%	0.867
Duration of 2nd-line antidiabetic (days)	0.458	1.580	0.034	500.73	58.0%	1.396